(17) A statement is said to be **contradiction** if its truth value is always $F$ irrespective of the truth values of its component statements. It is denoted by $\mathbb{F}$.

(18) A statement which is neither a tautology nor a contradiction is called **contingency**.

(19) Any two compound statements $A$ and $B$ are said to be **logically equivalent** or simply **equivalent** if the columns corresponding to $A$ and $B$ in the truth table have **identical truth values**. The logical equivalence of the statements $A$ and $B$ is denoted by $A \equiv B$ or $A \Leftrightarrow B$. Further note that if $A$ and $B$ are logically equivalent, then $A \leftrightarrow B$ must be a **tautology**.

(20) **Some laws of equivalence:**

**Idempotent Laws:** (i) $p \vee p \equiv p$ (ii) $p \wedge p \equiv p$.

**Commutative Laws:** (i) $p \vee q \equiv q \vee p$ (ii) $p \wedge q \equiv q \wedge p$.

**Associative Laws:** (i) $p \vee (q \vee r) \equiv (p \vee q) \vee r$ (ii) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.

**Distributive Laws:** (i) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

(ii) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

**Identity Laws:** (i) $p \vee \mathbb{T} \equiv \mathbb{T}$ and $p \vee \mathbb{F} \equiv p$

(ii) $p \wedge \mathbb{T} \equiv p$ and $p \wedge \mathbb{F} \equiv \mathbb{F}$

**Complement Laws :** (i) $p \vee \neg p \equiv \mathbb{T}$ and $p \wedge \neg p \equiv \mathbb{F}$

(ii) $\neg \mathbb{T} \equiv \mathbb{F}$ and $\neg \mathbb{F} \equiv \mathbb{T}$

**Involution Law or Double Negation Law:** $\neg(\neg p) \neg p$

**de Morgan's Laws:** (i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (ii) $\neg(p \vee q) \equiv \neg p \vee \neg q$

**Absorption Laws:** (i) $p \vee (p \wedge q) \equiv p$ (ii) $p \wedge (p \vee q) \equiv p$

## ICT CORNER

**https://ggbm.at/dy9kwgbt or Scan the QR Code**

Open the Browser, type the URL Link given (or) Scan the QR code. GeoGebra work book named "**12th Standard Mathematics Vol-2**" will open. In the left side of work book there are chapters related to your text book. Click on the chapter named "**Discrete Mathematics**". You can see several work sheets related to the chapter. Go through all the work sheets.

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